MILITARY SPECIFICATION SHEET

METER, READER FOR SOLID STATE ELAPSED TIME INDICATORS
BATTERY OPERATED, PORTABLE, HAND-HELD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The reader covered by this specification sheet shall be governed solely by this specification sheet. All requirements are specified here and take precedence over all requirements specified in MIL-M-7793.

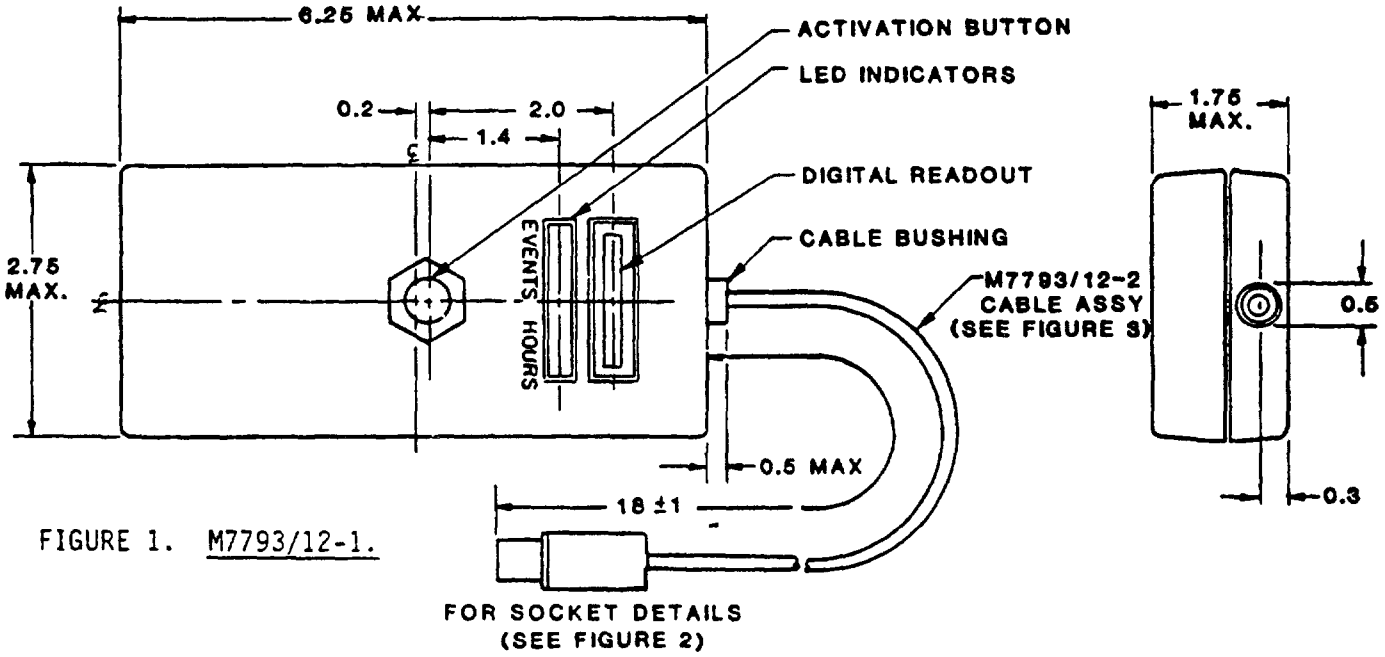


FIGURE 1. M7793/12-1.

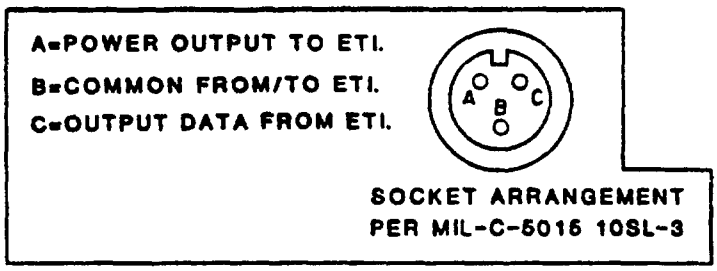


FIGURE 2. Reader socket assignment.

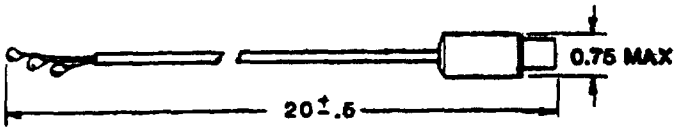


FIGURE 3. M7793/12-2 Replacement cable assembly.

AMSC N/A FSC 6645

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DESCRIPTION:

This reader is a hand-held battery-operated instrument which can power and read MIL-M-7793/13, /14, /15 and /16 solid state elapsed time indicators. The reader provides power to the reader input pin and extracts data from the output data pin of the solid state device being monitored. A third contact serves as a common. Data is transmitted over the M7793/12-2 cable and connector assembly. A single pushbutton switch initiates an automatic sequence of approximately 20 seconds duration during which the data is presented on a 7-digit light emitting diode display. At the completion of the sequence, the reader turns itself off.

REQUIREMENTS:

Dimensions: In accordance with Figure 1. Dimensions are in inches. Unless otherwise specified, tolerances are $\pm$0.1 for one place decimals.

Case of reader: Color shall be black for the M7793/12-1 reader.

Weight: Complete assembly, including battery, 15 oz. maximum.

Display: Display shall have seven light emitting diode (LED) digits and two light emitting diode status indicators. The status indicators show whether elapsed time or events are being displayed. For elapsed time indication, the decimal point will also be displayed. The digit height is 0.12 inches minimum. The display must be readable from 4 to 36 inches in any ambient light, varying from total darkness to high level office lighting. The display must be readable at angles up to 20 degrees from the display's perpendicular.

Durability: No mechanical or electrical degradation after 100 cycles. Operational accuracy shall be checked at end of test.

Temperature: Operational temperature range -20° to +55°C (see Note 8).

Storage temperature: Storage temperature range -55° to +85°C.

Altitude: Altitude (non-operational) 15,000 feet maximum.

Vibration: 10 to 55 Hz, 0.06 inch double amplitude, in accordance with MIL-STD-202, Method 201.

Shock: 50g peak, 11 ms saw tooth, in accordance with MIL-STD-202, Method 213, Test Condition G.

Operational accuracy: No deviation from actual elapsed time, as contained in the output data from a qualified MIL-M-7793/13-/16 solid state Elapsed Time Indicator (ETI).

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Battery: Power shall be supplied by one 9 volt NEDA (National Electronic Distributors Association) Type 1604 alkaline battery or military P/N BA-3090/U. Battery shall be installed when reader is shipped. The battery shall have a minimum life of 1200 readings or 2 years of calendar life, whichever occurs first, at 25°C. Low battery power shall be indicated by the display flashing with a two to one intensity modulation at a 4 Hertz rate during the 15 second display.

Power consumption: Discharge current shall not exceed 80 milliamperes any time during the operational cycle while reading any valid output data. Discharge current shall not exceed 2 microamperes when non-operational.

Data reading: The reader shall provide an accurate display of time when provided with a 32 bit serial word from the solid state elapsed time indicator on socket contact C. The reader must read a logical one of +3.3 to +6.6 volts and logical zero of 0 to +.2 volts from the elapsed time indicator. (Note: These are no-load voltages prior to reader attachment.)

Input voltage overload protection: The reader shall not be damaged by continuous input overloads of $\pm$50 volts or transient voltages of $\pm$500 volts for up to 10 microseconds between socket contacts B and C.

Power output: The reader shall provide a 9 $\pm$ 2 volt pulse at socket contact A (to power up ETI) for 3 $\pm$ 0.5 seconds, commencing within 5 seconds after the push-to-read button is depressed. The reader must not sustain damage if the output circuit is shorted between socket contacts A and B.

Connector: The connector at the end of the cable assembly shall be a MIL-C-5015 type modified with no threaded locking ring (modified MS3106A10SL-3S, or equivalent). The connector shall have an integral strain relief and mate with M7793/14-/16 elapsed time indicator connectors.

Test Sequence for Qualification, Quality Conformance Inspection and Retention of Qualification. This test sequence replaces the sequence of Section 4 of MIL-M-7793.

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Test	Qualification and Retention of Qualification	Quality Conformance Tests (100%)
Examination of Product	X	X
Power Consumption	X	X
Low Battery Power	X	
Input Voltage Overload	X	
Power Output	X	X
Weight	X	
Operational Accuracy	X	X
Altitude	X	
Salt Spray	X	
Vibration	X	
Mechanical Shock	X	
Moisture Resistance	X	
Life	X	

TEST METHOD:

Examination: The reader shall be inspected for all applicable dimensions, marking, proper color and weight conformance.

Durability: 100 cycles, tested in accordance with MIL-C-5015.

Power consumption: The reader shall be connected to a qualified MIL-M-7793/13-/16 solid state ETI or ETI simulator (Note 9) with a milliammeter connected in series with the battery. The battery current shall be measured before the reading cycle, during the cycle, and after completion of the cycle.

Low battery power: The reader shall be connected to a qualified MIL-M-7793/13-/16 solid state ETI. An adjustable DC power supply shall be substituted for the reader's battery. Set the power supply at 6.50 VDC and press the activation button; verify that the display does not flash. Reset the power supply at 6.10 VDC and press the activation button; verify that the display flashes.

Input voltage overload: An input voltage of +50 ± 2 volts shall be applied to the reader between socket contacts B and C for 1 minute. The voltage shall be reversed to -50 + 2 volts for 1 minute. An input voltage of +500 ± 20 volts shall be applied to the reader between sockets B and C for time less than 10 microseconds. The voltage shall be reversed to -500 ± 20 volts for the same duration. This may be simulated by use of a 1.0 ± .1 nanofarad capacitor charged to 500 ± 20 volts. Following the test, power output and operational accuracy shall be checked.

Power output: The reader shall be connected to a qualified MIL-M-7793/13-/16 solid state ETI with a voltmeter connected in parallel with contacts A and B (an adapter may be required). The output voltage shall stabilize at 9 ± 2 volts within 5 seconds of activation of the reader and remain at this level for 3 ± 0.5 seconds.

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Operational accuracy: The reader shall receive and correctly display the output data from a qualified MIL-M-7793/13-/16 solid state ETI or ETI simulator shown below. Timing shall be in accordance with the output data format of a qualified MIL-M-7793 ETI.

Output Data			Reader	
Decimal Equivalent	32 Bit Serial Word	Logical "1" Voltage	Digit Display	Status Indicator
12345.67	01 1110 0110 1010 0010 1100 0100 1000 01	+3.30	12345.67	Hours
12345.67	01 1110 0110 1010 0010 1100 0100 1000 01	+6.60	12345.67	Hours
76534.21	01 1000 0100 0010 1100 1010 0110 1110 01	+3.30	76534.21	Hours
76534.21	01 1000 0100 0010 1100 1010 0110 1110 01	+6.60	76534.21	Hours
00008.90	01 0000 1001 0001 0000 0000 0000 0000 01	+3.30	8.90	Hours
00008.90	01 0000 1001 0001 0000 0000 0000 0000 01	+6.60	8.90	Hours
00000.89	01 1001 0001 0000 0000 0000 0000 0000 01	+3.30	0.89	Hours
00000.89	01 1001 0001 0000 0000 0000 0000 0000 01	+6.60	0.89	Hours

Life: With a fresh battery installed, check power output and operational accuracy. Activate the reader for 1200 reading cycles (while reading a decimal equivalent of 12345.67 from a qualified MIL-M-7793/13-/16 solid state ETI or ETI simulator) at a repetition rate of approximately 4 minutes per cycle. At the conclusion of the test, check power output and operational accuracy. Test at room ambient conditions.

Moisture resistance: The reader assembly shall be subjected to Method 106 of MIL-STD-202. Initial measurements shall be power output and operational accuracy. No polarization or load voltages are to be applied. Final measurements shall be power output and operational accuracy after drying period.

Altitude: The reader assembly shall be subjected to Test Condition F of Method 105 of MIL-STD-202. The reader shall not be operated during the test. After the test, power output and operational accuracy shall be checked.

Salt spray: The reader assembly shall be subjected to Test Condition B of Method 101 of MIL-STD-202. The reader shall not be operated during the test. After the test, power output and operational accuracy shall be checked.

Vibration: The reader assembly shall be subjected to vibration in accordance with MIL-STD-202, Method 201. The reader shall be activated during the test once every 10 minutes. After the test, power output and operational accuracy shall be checked.

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Shock: The reader assembly shall be subjected to shock in accordance with MIL-STD-202, Method 213, Test Condition G. The reader shall not be operated during the test. After the test, power output and operational accuracy shall be checked.

Part numbers: The reader and cable/connector assembly shall be part number M7793/12-1. The repair cable/connector shall be part number M7793/12-2.

NOTES:

Operation

1. The reader's connector shall be connected to the elapsed time indicator's (ETI) or counter's connector by the M7793/12-2 cable assembly. Connectors mate by pushing together.

2. The push-to-read button of the reader shall be depressed, then released.

3. The reader will activate the ETI within 5 seconds of pushing the button on the reader. Data transmission is confirmed by a 2 Hertz alternate lighting of the LED status indicators.

4. The elapsed time will be displayed for 15 $\pm$ 5 seconds along with the correct status LED.

5. If an error is detected, the display will show 7 dashes for 15 $\pm$ 5 seconds.

6. A flashing display (see battery requirement) indicates a low battery. The battery is replaced by removing the 6 screws on the rear case half and unplugging the battery.

7. Cable/connector assembly replacement: Remove the 6 screws on the rear case half. Unscrew three ring terminals. Remove RTV adhesive-sealant from within the cable bushing and withdraw the old cable assembly. Install the new M7793/12-2 assembly. Tighten all screws and reseal the cable bushing with RTV (RTV 103 or equivalent).

8. The reader may be able to be operated intermittently over an extended temperature range of -40°C to 71°C, but operation is dependent on the battery's characteristics and charge state.

9. An ETI simulator that has been approved by the qualifying activity may be used in lieu of a qualified MIL-M-7793/13-/16 solid state ETI.

Custodian:
Navy - AS
Army -
Air Force - 99

Users:
Navy - SH, OS

Reviewer:
Army - MI

Preparing activity:
Navy - AS
(Project Number 6645-0399)

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